# Homework 5

Name Here

## A model for bee toxicity

Again consider the bee toxicity problem from the previous homework. The data on toxicity levels and hive collapse is given again below for convince.

| x | y |
|---|---|
| 1.06 | 0 |
| 1.41 | 1 |
| 1.85 | 1 |
| 1.50 | 1 |
| 0.46 | 0 |
| 1.21 | 1 |
| 1.25 | 1 |
| 1.09 | 1 |
| 1.76 | 1 |
| 1.75 | 1 |

Assume that beehive collapse, $y_i$, given pollutant exposure level $x_i$, is $Y_i \sim \text{Bernoulli}(\theta(x_i))$, where $\theta(x_i)$ is the probability of death given dosage $x_i$. We will assume the *probit* regression model, $\theta_i(x_i) = \Phi(\alpha + \beta x_i)$ where $\Phi$ denotes the CDF of a standard normal distribution. Probit regression has a simple latent variable interpretation that is particularly amenable to Gibbs sampling. We will implement the Gibbs sampler by deriving the complete conditionals in a model where we have added additional augmented paramters, $z_i$.

**1a** Let $z_i \sim N(\alpha + \beta x_i, 1)$. Show that $y_i = I[z_i > 0]$ (1 if $z_i$ is greater than 0, and 0 otherwise) implies the probit model above.

**1b**. Derive the complete conditionals $p(\alpha \mid z, y, x, \beta)$ and $p(\beta \mid z, y, x, \alpha)$ assuming a flat prior on $\alpha$ and $\beta$.

**1c** Now derive the complete conditionals of $\alpha$ and $\beta$ assuming instead assuming a priori $\alpha \sim N(-10, \sigma = 5)$ and $\sim N(0, \sigma = 10)$ (note: standard deviations are provided not variances).

**1d** Derive the complete conditional of $z_i$, $p(z_i \mid z_{-i}, y, x, \alpha, \beta)$. Find an R package or function that you can use to sample from this density and report the package you plan to use.

**1e**. Use the above conditionals, assuming the normal priors on $\alpha$ and $\beta$, and implement and run a Gibbs sampler with multiple chains. Run the sampler long enough so that the effective sample size for all parameters is at least 1000 and discuss mixing. Report at 95% confidence interval for the LD50 and report the posterior probability that LD50 is greater than 1.2.

### Stochastic volatility models

Stochastic Volatility (SV) models, are a staple in financial econometrics. Unlike standard regression models where variance is constant (homoscedasticity), SV models assume the variance itself is a latent process that evolves over time. While conceptually simple, SV models are notorious to sample because of the posterior geometry. Specifically, they often exhibit "funnel" shapes that cause Hamiltonian Monte Carlo (HMC) samplers to fail.

Let $y_t$ be the percentage return of an asset at time $t$. We model $y_t$ with a latent log-volatility state $h_t$:

$$y_t \sim \mathcal{N}(0, \exp(h_t/2))$$
$$h_t \sim \mathcal{N}(\mu + \phi(h_{t-1} - \mu), \sigma)$$

Where $\mu$ is the mean log-volatility, $\phi$ is the persistence of the volatility $(-1 < \phi < 1)$ (the autocorrelation parameter) and $\sigma$ is the volatility of volatility (how "jumpy" the risk is).

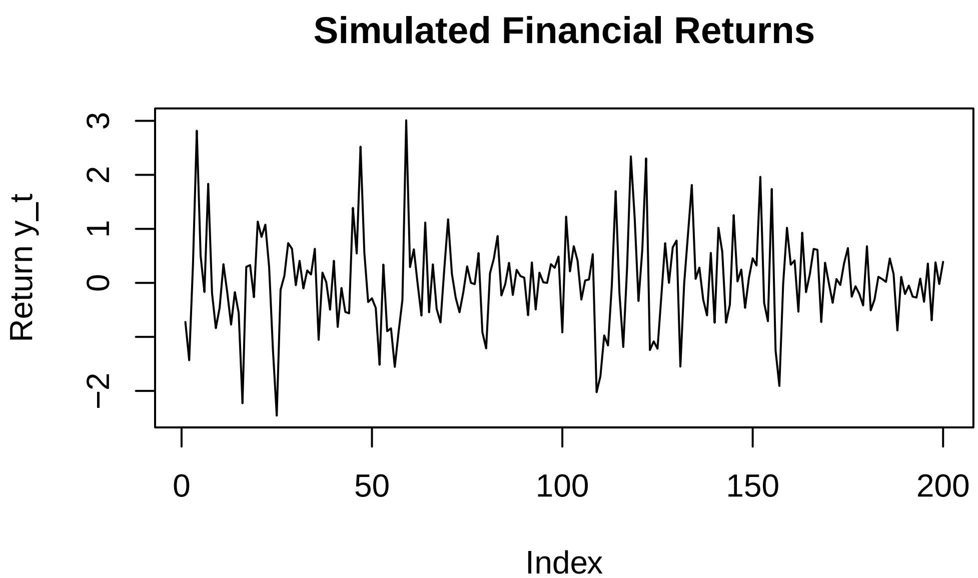
Below we generate a synthetic dataset where we know the true parameters.

```r
set.seed(215)
T_obs <- 200
mu_true <- -1.0
phi_true <- 0.95
sigma_true <- 0.5


h_true <- numeric(T_obs)
h_true[1] <- rnorm(1, mu_true, sigma_true / sqrt(1 - phi_true^2))
for (t in 2:T_obs) {
  h_true[t] <- rnorm(1, mu_true + phi_true * (h_true[t-1] - mu_true), sigma_true)
}

# Simulate observed returns
y <- rnorm(T_obs, 0, exp(h_true / 2))
```

```r
# Visualize
plot(y, type = 'l', main = "Simulated Financial Returns", ylab = "Return y_t")
```




**2a.** A `stan` file to sample from this model is included in `cv_centered.stan`. Please look over the code and make sure you understand it. Run four chains of the sampler with the default number of warmpu and sample iterations (1000 for each). Report the following statistics:

- The posterior mean of $\sigma$ and $\phi$
- The effective sample sizes for $\sigma$ and $\phi$
- Any parameters with Rhats $> 1.1$
- The number of divergent transitions (if any)

**2b.** To improve the sampler, reimplement the model using a "Non-Centered Parameterization (NCP)". To fix the issues you saw in the previous fit, we need to break the dependency between $h$ and $\sigma$ in the prior. We do this by sampling a standard normal variable h_std (which has no unknown parameters in its distribution) and transforming it to get $h$. Create a new stan file that implements the model using NCP. You should define `h` in the "transformed parameters" block as a function of standard normal random variables, as well as $\mu$ and $\phi$.

With the samples generated using the non-centered parameterization, again report:

- The posterior mean of $\sigma$ and $\phi$
- The effective sample sizes for $\sigma$ and $\phi$
- Any parameters with Rhats $> 1.1$
- The number of divergent transitions (if any)

Note that you should see that the first 3 metrics improve but you may find some divergent transitions (unlike with the centered parameterization.) If you see divergent transitions, one option is to increase the value of `adapt_delta` in the cmdstan sample argument. This parameter controls the target acceptance probability of the NUTS. By default it is set to 0.8. Higher acceptance rates imply smaller stepsizes in the leapfrog integrator, which should reduce divergences. Set `adapt_delta=0.99`. You should notice that the sampler takes longer to run, but you should also see a reduction in divergences.